

HIGH SPEED VCO SWEEPER

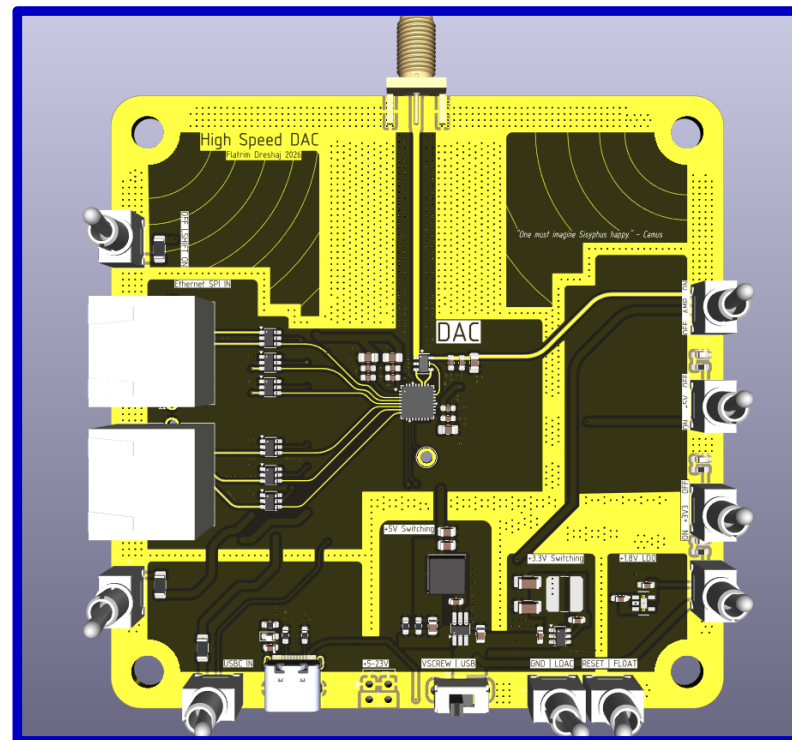
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Rev: 0

Date: 2026-08-27

F. Dreshaj

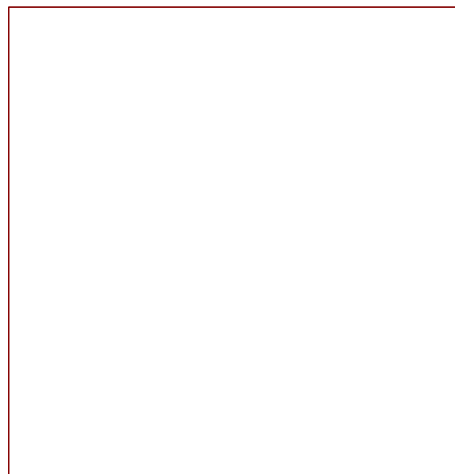


Research and Simulation [1]



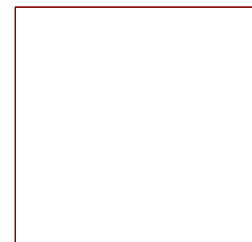
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AD3551R Board [2]



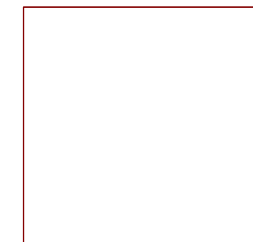
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Assembly and Testing [6]



File: a_and_t.kicad_sch

Results [7]



File: Results.kicad_sch

Flatrim Dreshaj

Sheet: /

File: AD3551R_DAC.kicad_sch

Title: DAC Sweep

Size: A4

Date: 2026-08-22

KiCad E.D.A. 10.0.5

Rev: 0

Id: 0/8

1	2	3	4	5	6
A					A
B					B
C					C
D					D
1	2	3	4	5	6

Sheet: /Research and Simulation [1]/ File: r_s.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 1/8

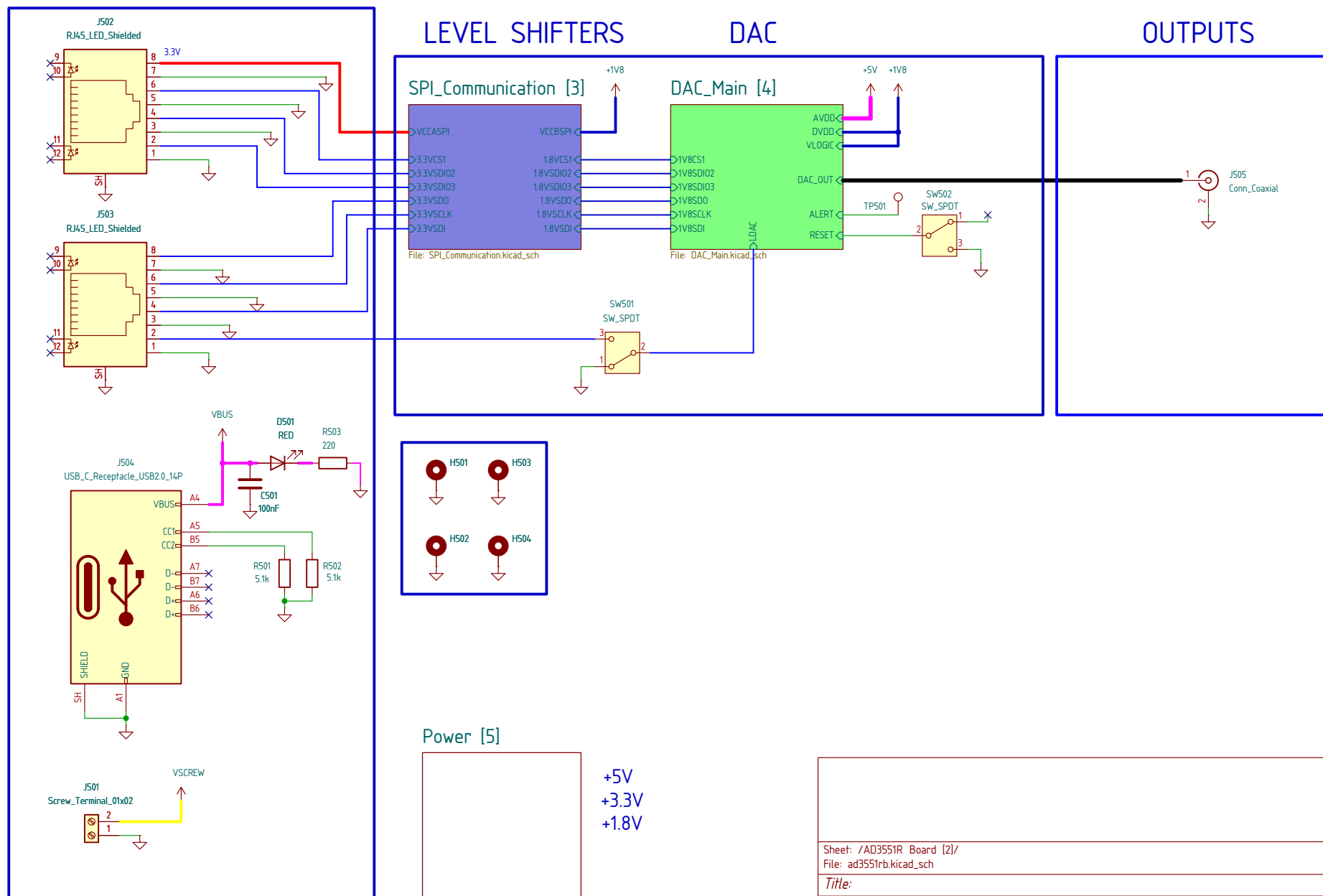
AD3551R DAC Fast Sweep Board [2]

INPUTS

LEVEL SHIFTERS

DAC

OUTPUTS



Power [5]

+5V
+3.3V
+1.8V

Sheet: /AD3551R Board [2]/
File: ad3551rb.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.5

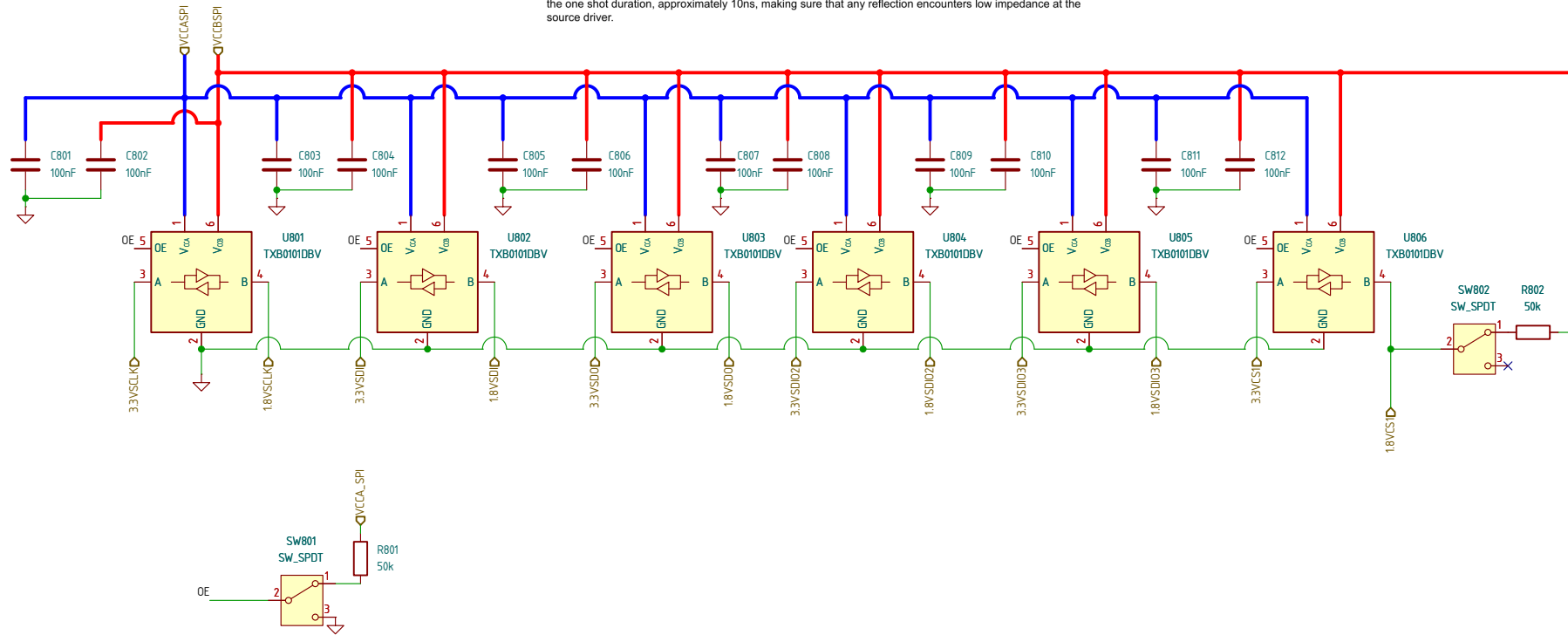
Rev:

Id: 2/8

7.4.1 Layout Guidelines

For device reliability, the following common printed-circuit board layout guidelines are recommended.

- Bypass capacitors should be used on power supplies, and should be placed as close as possible to the V_{CCA} , V_{CCB} pin and GND pin.
- Short trace lengths should be used to avoid excessive loading.
- PCB signal trace-lengths must be kept short enough so that the round-trip delay of any reflection is less than the one shot duration, approximately 10ns, making sure that any reflection encounters low impedance at the source driver.



Sheet: /AD3551R Board [2]/SPI_Communication [3]/
File: SPI_Communication.kicad_sch

Title:

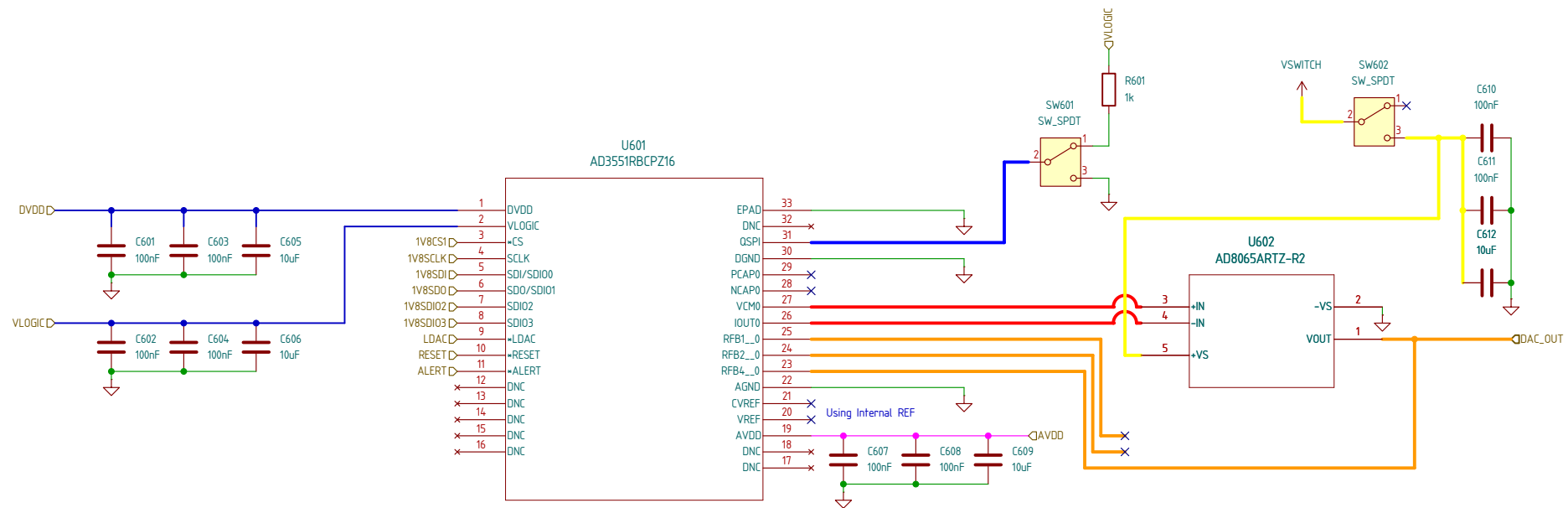
Size: A4

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Layout Notes:

QFN Package Extend pads

I_Out as short and thin as possible due to amplifier slew.

Route Line with constant characteristic impedance for timing

Sheet: /AD3551R Board [2]/DAC_Main [4]/
File: DAC_Main.kicad_sch

Title:

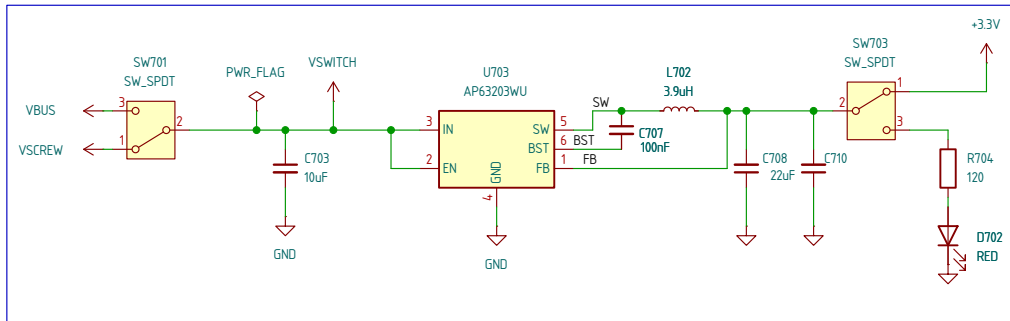
Size: A4

Date:

Rev:

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Id: 4/8



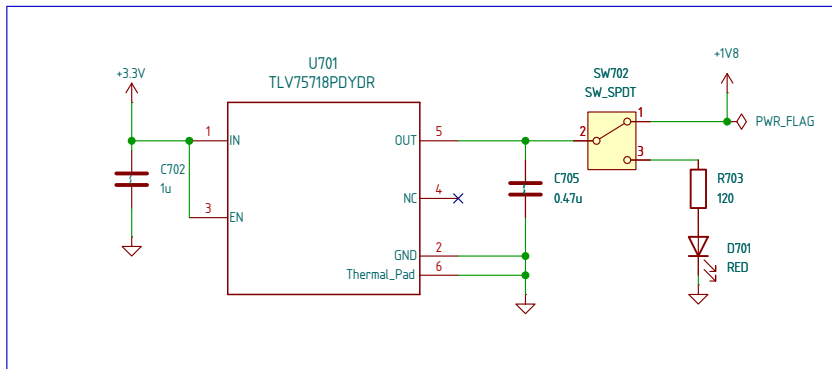
Buck: 3.3V 2A

Power Consumption Chain

USB-C - 500mA
Screw Terminal - 16A

5V Switching - 2A
3.3V Switching - 2A
1.8V LDO - 1A

AD3551 DAC - 60.875mA see table below
AD8065 OpAmp - 14.4mA see table below



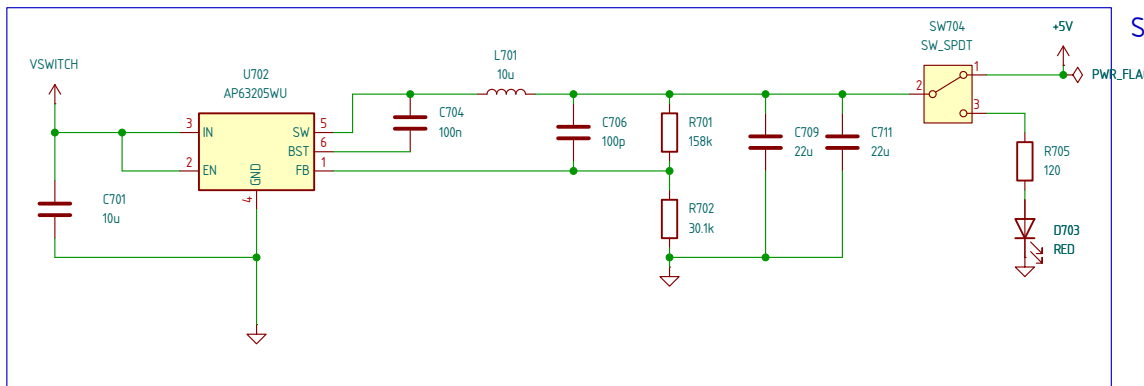
LDO +1.8V 1A

Analog 5V Rail Current Consumption

Block	Typ Current (mA)	Max Current (mA)	Typ Power (mW)	Max Power (mW)
AD3551R DAC AVDD	12.0	15.0	125.702	157.454
AD8065	12.8	14.4	64.000	72.000
5V RAIL TOTAL	24.8	29.4	189.702	189.702

Digital 1.8V Rail Current Consumption

VLOGIC Pin section		Voltage (V)	Typ Current (mA)	Max Current (mA)	Typ Power (mW)	Max Power (mW)
VLOGIC Current	ILOGIC	1.8	0.001	0.0075	0.002	0.014
VLOGIC Dyn Current	ILOGIC_DYNAMIC	1.8	3.0	5.0	5.400	9.000
DVDD Pin section						
DVDD Current	IDVDD	1.8	0.5	0.8	0.900	1.440
DVDD Dynamic	IDVDD_DYNAMIC	1.8	33.0	40.0	59.400	72.000
Current				45.875		82.454



Switching: 5V 2A

Sheet: /AD3551R Board [2]/Power [5]/
File: Power.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.5

Rev:

Id: 5/8

	1	2	3	4	5	6
A						
B						
C						
D						
	1	2	3	4	5	6

Sheet: /Assembly and Testing [6]/ File: a_and_f.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 6/8

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B					B
C					C
D					D
1	2	3	4	5	6

Sheet: /Results [7]/ File: Results.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 7/8